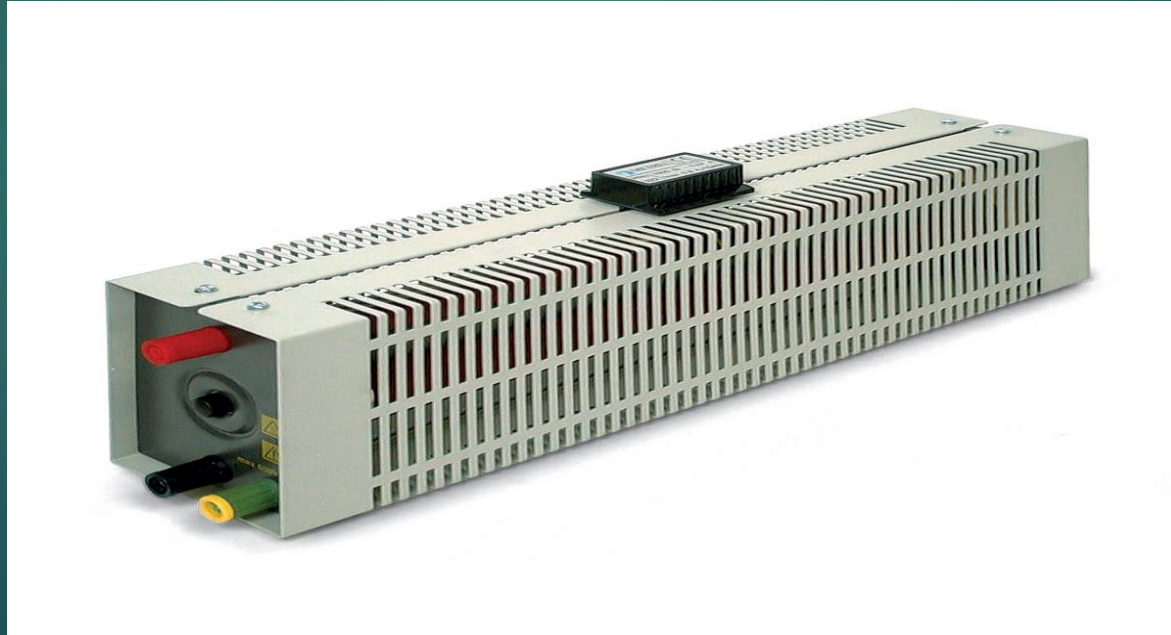


Experiment #(8)

Potentiometer

مجزء الجهد



Objective

- ▶ To build up a Potentiometer using a conducting wire.
- ▶ To determine the resistivity of a conducting wire.
- ▶ To determine the electromotive force (*emf*) of a battery.

Apparatus

Power Supply



Slide Wire



Voltmeter



Galvanometer



Battery



Banana Wires



Theory

In the circuit shown in figure 1. The reading of the voltmeter is given by the following relation:

$$V = IR = I \frac{\rho L}{A} \dots\dots\dots(1)$$

Where ρ is the resistivity in ($\Omega \cdot m$), L is the Length in (m) and A is the area cross-section in (m^2).

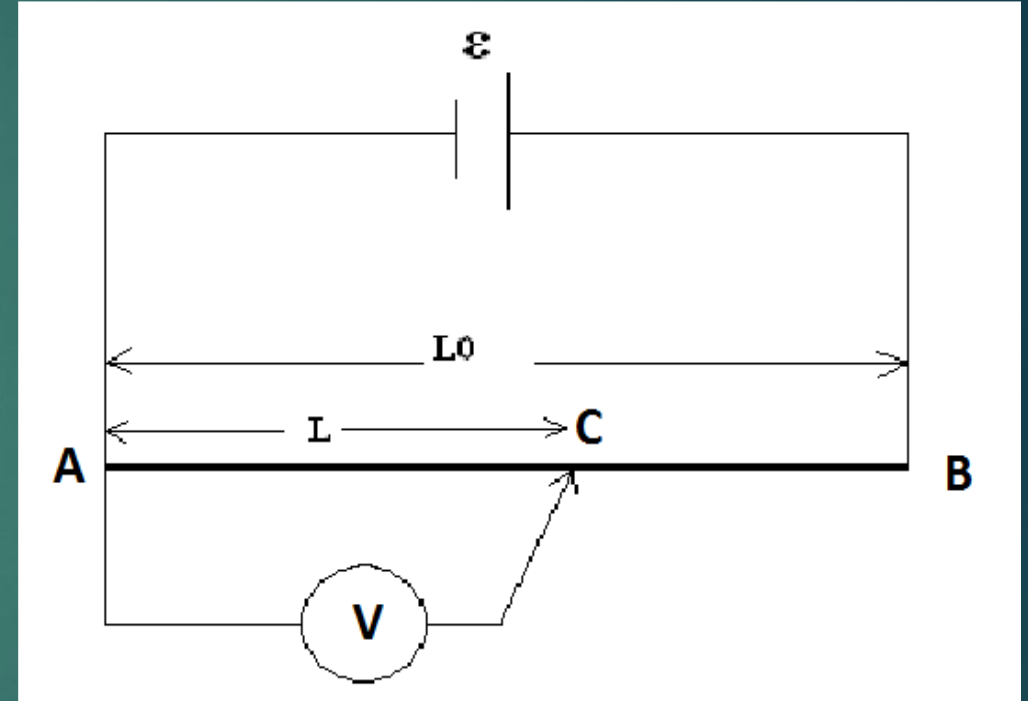


Figure 1

If we replace the voltmeter in figure 1 by a battery of unknown emf ϵ_x and a galvanometer as shown in figure 2.

$$V = \epsilon_x \pm I_G (r + R_G) \dots\dots\dots (2)$$

Where R_G is the resistance of the galvanometer in (Ω), I_G is the current through the galvanometer in (amp) and r is internal resistance of the battery in (Ω)

If the galvanometer reads zero at a point at length L_x on the slide wire. Then

$$V = \epsilon_x$$

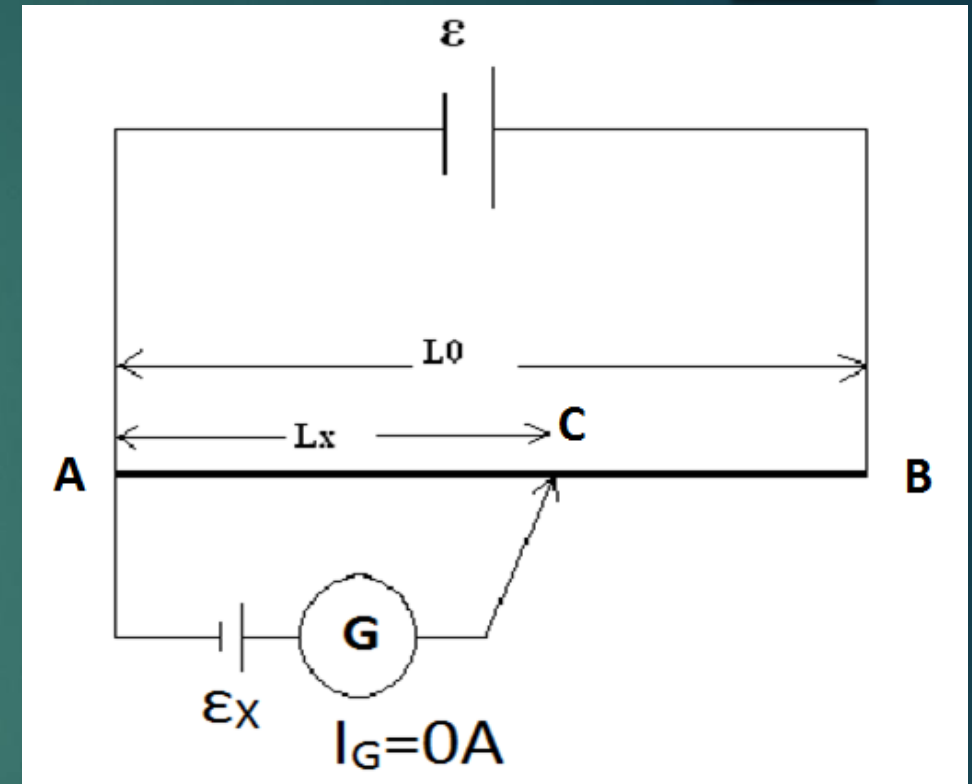


Figure 2

Procedure

1. Connect the circuit shown in Figure 1. Have your lab instructor check it before you make the connection to the power supply.
2. Vary the position of the sliding tap slightly at any desired point along the length of the slide wire.
3. Record the reading of the voltmeter with the length L in table 1. (Five trials at least).
4. Draw the relation between V versus L on the graph and calculate the slope.
5. Measure the diameter of the slide wire to calculate the area cross-section.
6. Use the slope to calculate the resistivity of the slide wire.
7. Connect the circuit shown in Figure 2. Have your lab instructor check it before you make the connection to the power supply.
8. Vary the position of the sliding tap slightly to balance at any desired point along the length of the slide wire and find the length L_x (the galvanometer reads zero).
9. From the graph find ϵ_x .



THANK YOU

